

Further Mechanics 1

Notes · Edexcel A Level Further Mathematics — Further Mechanics 1 · Chapter 5: Elastic Collisions in Two Dimensions

Chapter 5: Elastic Collisions in Two Dimensions

Why this chapter matters

Why this matters. This is the capstone of the unit: oblique impacts combine everything so far — momentum, restitution, and careful vector resolution — into genuinely two-dimensional problems. The entire technique rests on one idea: split every velocity into a component along the line of centres and a component perpendicular to it, then treat each direction with the right rule.

Key vocabulary

Term	Meaning
Line of centres	The line joining the centres of two colliding spheres at the moment of impact (for a sphere hitting a flat surface, this is the normal to the surface at the point of contact)
Oblique impact	A collision in which the velocity of at least one particle is not directed along the line of centres
Smooth sphere / smooth surface	“Smooth” means no friction acts during impact, so there is no force, and hence no impulse, in the direction perpendicular to the line of centres
Angle of deflection	The angle through which a sphere's direction of travel turns as a result of an oblique impact

Key formula table

Quantity	Formula / rule
Perpendicular to line of centres	component of velocity unchanged — true for each sphere individually, and for a sphere striking a fixed surface, because the surface is smooth
Along the line of centres — sphere and fixed surface	$v = eu$ — speed along the normal reverses and scales by e ; component perpendicular is unchanged
Along the line of centres — two spheres	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ (components along line of centres only) — exactly as in 1D, but applied only to the along-line-of-centres components

Newton's law of restitution (along line of centres)	$\mathbf{v}_2 - \mathbf{v}_1 = e(\mathbf{u}_1 - \mathbf{u}_2)$ — again, components along the line of centres only
Resultant speed after impact	$v = \sqrt{v_{\parallel}^2 + v_{\perp}^2}$ — combine the two resolved components back into a resultant
Direction after impact	$\tan\theta = v_{\perp} / v_{\parallel}$ — θ measured from the line of centres

Concept: Oblique impact of a smooth sphere with a fixed surface

When a smooth sphere strikes a fixed plane surface obliquely, resolve the velocity into two components: **along the normal to the surface** (the line of centres direction) and **parallel to the surface**. Because the surface is smooth:

- **Perpendicular** to the surface is unaffected by the impact (no friction $\Rightarrow$ no impulse in that direction).
- **Along the normal** is affected exactly as in a 1D wall collision: it reverses direction and its magnitude scales by e (speed after = $e \times$ speed before, along the normal).

Physical intuition. Picture hitting a ball with a bat at an angle: only the push directly into the ball — along the line of centres — changes how fast it moves towards or away from the bat. The sideways sliding motion continues completely untouched, because with no friction there is nothing to grip the surface and alter it. That is the whole chapter in one picture: split the motion into “the part that gets hit” and “the part that doesn't.”

Worked example 1 — sphere striking a fixed wall obliquely

EXAMPLE 5.1

Question. A smooth sphere strikes a fixed vertical wall with speed 10 m s^{-1} , at an angle of 40° to the wall. The coefficient of restitution between the sphere and the wall is 0.7 . Find the speed and direction of the sphere immediately after impact.

Step 1. The line of centres is perpendicular (normal) to the wall. The sphere's velocity is given at 40° to the wall itself, so resolve directly using that angle: the component **along the normal** (perpendicular to the wall) is found with $\sin 40^\circ$, and the component **along the wall** (parallel to it) with $\cos 40^\circ$ — there is no need to convert to the angle from the normal (50°) first, since $\sin$ and $\cos$ of complementary angles already do that swap for you:

- Component along the normal (towards wall): $u_n = 10 \sin 40^\circ = 6.428 \text{ m s}^{-1}$ **M1A1**
- Component along the wall (parallel): $u_p = 10 \cos 40^\circ = 7.660 \text{ m s}^{-1}$ **A1**

Step 2. Component along the wall is unchanged by the smooth impact: $v_p = 7.660 \text{ m s}^{-1}$. **B1**

Step 3. Component along the normal reverses and scales by e : $v_n = e \times u_n = 0.7 \times 6.428 = 4.499 \text{ m s}^{-1}$ (directed away from the wall). **M1A1**

Step 4. Recombine to find the resultant speed:

$$v = \sqrt{v_n^2 + v_p^2} = \sqrt{(4.499^2 + 7.660^2)} = \sqrt{(20.24 + 58.68)} = \sqrt{78.92} = 8.884 \text{ m s}^{-1} \text{ (3 s.f.)} \quad \text{M1A1}$$

Step 5. Direction: angle to the wall, ϕ , satisfies $\tan\phi = v_n/v_p = 4.499/7.660 \Rightarrow \phi = 30.4^\circ$ (3 s.f.)

M1A1

Answer: the sphere rebounds at approximately 8.88 m s^{-1} , at about 30.4° to the wall.

EXAM TIP

The single biggest source of error in this whole chapter is resolving the wrong component along the wrong direction — it is easy to reverse both components, or reverse the parallel one by mistake. Learn the rule as a mantra: “perpendicular to the line of centres, nothing changes; along the line of centres, it’s a 1D restitution problem.” A clear diagram with the incoming and outgoing velocity triangles, both split into the same two perpendicular directions, prevents almost every mistake here.

Concept: Oblique impact of two smooth spheres

When two smooth spheres of equal or unequal radii collide obliquely, the line of centres joins the two centres at the instant of impact. Resolve **each** sphere’s velocity into a component along the line of centres and a component perpendicular to it:

- **Perpendicular to the line of centres:** each sphere’s component is unchanged (no friction acts).
- **Along the line of centres:** apply conservation of momentum and Newton’s law of restitution to the along-line-of-centres components only, exactly as in a 1D collision (Chapter 4).

This means a 2D oblique collision problem reduces to one 1D collision calculation (for the line-of-centres components) plus two “components unchanged” statements (for the perpendicular components), followed by recombining each sphere’s two components back into a resultant speed and direction.

Worked example 2 — oblique collision between two spheres, one initially at rest
EXAMPLE 5.2

Question. Two smooth spheres, P of mass 2 kg and Q of mass 3 kg, collide. Immediately before impact, P is moving with speed 6 m s^{-1} at 30° to the line of centres, and Q is at rest. The coefficient of restitution between the spheres is 0.6. Find the speed and direction of each sphere immediately after impact.

Step 1. Resolve P’s initial velocity along and perpendicular to the line of centres:

- Along line of centres: $u_P = 6 \cos 30^\circ = 5.196 \text{ m s}^{-1}$ **M1A1**
- Perpendicular: $w_P = 6 \sin 30^\circ = 3 \text{ m s}^{-1}$ **A1**

Q is at rest, so both its components before impact are zero: $u_Q = 0$, $w_Q = 0$.

Step 2. Perpendicular components are unchanged for both spheres (smooth impact):

$$w_P (\text{after}) = 3 \text{ m s}^{-1}, w_Q (\text{after}) = 0 \text{ m s}^{-1} \quad \text{B1}$$

Step 3. Along the line of centres, treat as a 1D collision. Conservation of momentum:

$$2(5.196) + 3(0) = 2v_P + 3v_Q \quad \text{M1}$$

$$10.392 = 2v_P + 3v_Q \dots (1) \quad \text{A1}$$

Step 4. Newton's law of restitution along the line of centres:

$$v_Q - v_P = 0.6(5.196 - 0) = 3.118 \dots (2) \quad \text{M1A1}$$

Step 5. Solve (1) and (2) simultaneously. From (2): $v_Q = v_P + 3.118$. Substitute into (1):

$$10.392 = 2v_P + 3(v_P + 3.118) = 5v_P + 9.353 \quad \text{M1}$$

$$1.039 = 5v_P \Rightarrow v_P = 0.2078 \text{ m s}^{-1}, \text{ then } v_Q = 3.326 \text{ m s}^{-1} \quad \text{A1}$$

Step 6. Recombine each sphere's components. For P: along-line-of-centres component 0.208 m s^{-1} , perpendicular component 3 m s^{-1} :

$$\text{speed of P} = \sqrt{(0.208^2 + 3^2)} = \sqrt{9.043} = 3.007 \text{ m s}^{-1} \text{ (3 s.f.)} \quad \text{M1A1}$$

$$\text{angle to line of centres: } \tan \theta_P = 3/0.208 \Rightarrow \theta_P = 86.0^\circ \text{ (3 s.f.)} \quad \text{A1}$$

For Q: only a line-of-centres component (perpendicular component is zero), so Q moves at 3.33 m s^{-1} (3 s.f.) **along the line of centres.** **A1**

Step 7 (extension — energy check). As in Chapter 4, KE lost = total KE before – total KE after. Before: $\frac{1}{2}(2)(6^2) = 36 \text{ J}$ (Q is at rest). After: $\frac{1}{2}(2)(3.007^2) + \frac{1}{2}(3)(3.326^2) = 9.04 + 16.6 = 25.6 \text{ J}$ (3 s.f.). **M1**

KE lost $\approx 36 - 25.6 = 10.4 \text{ J}$ (3 s.f.). **A1** As expected, since $e = 0.6 < 1$, energy is lost — only for $e = 1$ would this oblique collision conserve KE.

Answer: P moves off at approximately 3.01 m s^{-1} at about 86.0° to the line of centres (i.e. sharply deflected, almost perpendicular to its original line-of-centres direction); Q moves along the line of centres at approximately 3.33 m s^{-1} .

EXAM TIP

This example is worth pausing on to build intuition: P's perpendicular component (3 m s^{-1}) barely changes, but its along-line-of-centres component drops dramatically (from 5.2 to 0.2 m s^{-1}) because most of P's “approach” momentum along that line transfers to Q. The result — P deflected through a large angle — is physically exactly what you'd expect from an off-centre hit, and connecting the algebra to that everyday intuition (a glancing blow deflects the mover sharply while barely slowing their sideways drift) helps you remember which component does what, rather than just grinding through formulae.

Chapter summary

Term	Meaning
Setting up	always resolve along and perpendicular to the line of centres before doing anything else
Perpendicular to line of centres	unchanged for each sphere/for the sphere in a wall collision (smooth $\Rightarrow$ no friction impulse)

Along line of centres, sphere vs fixed surface	$v = eu$, direction reverses
Along line of centres, two spheres	ordinary 1D collision: conservation of momentum + Newton's law of restitution
Recombining	speed = $\sqrt{(\text{parallel}^2 + \text{perpendicular}^2)}$; direction from $\tan\theta = \text{perpendicular/parallel}$, measured from the line of centres

Common exam mistakes

- Resolving velocities relative to the wrong reference line — always the line of centres (or the normal to a flat surface), never the direction of initial motion.
- Applying Newton's law of restitution or conservation of momentum to full (unresolved) velocities instead of only the along-line-of-centres components.
- Forgetting to check whether the perpendicular component should be zero (e.g. a sphere at rest, or a sphere moving exactly along the line of centres) before setting up unnecessary algebra.
- Losing track of which angle (to the line of centres, or to the surface) is being asked for, and quoting an angle measured from the wrong reference line.
- Arithmetic slips when recombining components with Pythagoras and inverse tan — always sanity-check the direction against the diagram (e.g. a sphere deflected through a large angle should look — and be labelled — like a glancing blow).

You should now be able to

- Resolve a sphere's velocity into components along and perpendicular to the line of centres for an oblique impact.
- Apply the rule that the perpendicular component is unchanged in a smooth collision.
- Apply $v = eu$ along the normal for a sphere striking a fixed smooth surface, and recombine components to find speed and direction after impact.
- Apply conservation of momentum and Newton's law of restitution along the line of centres for an oblique collision between two smooth spheres, and recombine components for each sphere afterwards.